WaveHFG: High-Frequency Guidance for Heterogeneous Remote Sensing Image Change Detection With Wavelet Features

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***Abstract*—Heterogeneous change detection (Hete-CD) between optical and synthetic aperture radar (SAR) images integrates detailed spectral information with all-weather observation capabilities. This approach aims to address the limitations of optical images, such as cloud cover and illumination variations, while mitigating speckle noise and enhancing the interpretability of SAR imagery. However, integrating these modalities poses challenges, including spectral inconsistencies and mismatched feature representations. To overcome these challenges, we propose a wavelet high-frequency guidance (WaveHFG) change detection (CD) network. This approach utilizes wavelet-transform high-frequency features to enhance both the similarity and directional consistency of representations extracted from heterogeneous images. Our method incorporates two key modules: high-frequency differential-guidance (Diff-G) and high-frequency directional-guidance (Dir-G). These modules effectively capture subtle and often-overlooked details, hence improving the interpretability of the results. Additionally, the frequency–spatial domain difference fusion (FSD^2F) module integrates features across multiple domains, providing a more comprehensive and detailed representation of change information. To rigorously evaluate the effectiveness of our proposed method, we constructed a new Hete-CD dataset with extensive coverage and increased complexity, encompassing a broader range of target categories to better reflect diverse real-world conditions. Extensive experiments on two publicly available datasets and our newly proposed dataset, demonstrate that our method outperforms state-of-the-art CD methods. Both the source code and the dataset are available at https://github.com/songxy9037/WaveHFG**

***Index Terms*—Change detection (CD), convolutional neural network, deep learning, frequency domain, heterogeneous images, wavelet transform.**

I. Introduction

REMOTE sensing image processing plays a crucial role in Earth observation, such as environmental monitoring [1], urban planning [2], agricultural management [3], and disaster assessment [4]. Advances in sensor technology have enabled collaborative observation using multimodal data, collectively referred to as heterogeneous remote sensing imagery (HRSI). Analyzing HRSI delivers richer and more detailed information, thus improving the adaptability of change detection (CD) across diverse scenes.

Optical images, owing to their intuitive visual properties and rich spectral information, have long been the primary data source for CD tasks [5], [6], [7], [8], [9]. However, optical images are difficult to acquire under conditions such as nighttime or heavy cloud cover, which limits their real-time responsiveness in emergent scenarios. In such cases, heterogeneous images from other sensors provide critical support for addressing these limitation.

Traditional CD techniques, such as differential analysis [10], threshold segmentation [11], principal component analysis (PCA) [12], clustering [13], and change vector analysis (CVA) [14], rely on fundamental image processing and statistical methods. Compared to complex deep learning models, these methods typically exhibit lower computational complexity, and are suitable for resource-limited environments. However, their effectiveness diminishes when applied to heterogeneous images, where variations in spectral, spatial, and temporal resolutions pose significant challenges. Specifically, techniques like differential analysis and thresholding struggle to accurately distinguish changes across diverse image pairs, as these methods are sensitive to resolution disparities that hinder precise segmentation and detection. To address these limitations, recent hybrid methods [15], [16] have emerged, combining traditional feature extraction with machine learning techniques to improve heterogeneous CD (Hete-CD) performance.

In recent years, researchers have increasingly explored Hete-CD methods [17], [18] based on deep neural networks. Deep learning models can automatically learn high-level features, significantly enhancing the accuracy and robustness of CD. The simplest deep learning method [19], [20] involves inputting bitemporal heterogeneous images into a Siamese neural network. This network employs a shared-weight encoder to extract features, enhancing the similarity between bitemporal features and hence improving CD performance. However, this method heavily depends on the network's ability to represent features and often fails to accurately capture distinguishing features when the modalities are insufficiently similar.

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To address the issue of modal heterogeneity, researchers have proposed some methods for feature matching between different modalities. Two primary approaches are commonly employed: one maps features from different modalities into a unified feature space to reduce discrepancies and enhance feature comparability [18], [21], [22]; the other converts the features of one modality into those of another. Among them, generative adversarial networks (GANs) [23], [24], [25] have been widely used to learn mappings between multimodal features.

Compared to Siamese strategies that rely solely on deep feature extraction, modality-matching approaches are more suitable for Hete-CD. However, these matching methods still face a fundamental challenge. They attempt to directly compare features of heterogeneous images in the spatial domain. This direct comparison is difficult because significant low-frequency differences, such as brightness and color. This can cause the network to favor one image type over the other, leading to inaccurate results. As a result, these methods struggle to robustly capture the shared high-frequency information, like fine details and sharp edges, especially in noisy synthetic aperture radar (SAR) images.

Beyond these algorithmic hurdles, the field is also constrained by the limitations of existing public datasets. While benchmarks like the U.K.-HCD and France-HCD datasets are valuable, they offer a very limited number of samples for training modern deep learning models. Furthermore, their thematic scope is narrow, each focusing on a single event type (e.g., flooding or construction). This scarcity of large-scale and diverse datasets creates a significant bottleneck, hindering the development of truly robust Hete-CD models.

To address these dual challenges, we propose solutions for both algorithms and data. On the algorithm side, we shift the focus from the spatial domain to the frequency domain using a wavelet transform. This approach is designed to overcome the core problem of spatial-domain methods: the instability caused by large low-frequency differences. By separating frequency components, our method can focus on the shared and stable high-frequency information (e.g., edges and textures), which provides a more robust basis for CD. On the data side, to mitigate the issue of data scarcity, we construct a new dataset for Hete-CD. It incorporates both optical and SAR images, offering a more challenging and representative benchmark.

In this article, we propose wavelet high-frequency guidance (WaveHFG) Network, a novel architecture for Hete-CD. Leveraging the high-frequency similarity of wavelet transform in heterogeneous image processing, we address the cross-domain matching issue. Fig. 1 illustrates the process of generating a change map using high-frequency features from wavelet transforms. The network follows an encoder–decoder structure and consists of three core modules: high-frequency differential guidance (Diff-G), high-frequency directional guidance (Dir-G), and frequency–spatial domain difference fusion (FSD^{2}F). First, an improved encoder serves as the backbone, extracting multilevel features. These features are then decomposed into high-frequency and low-frequency components using wavelet transform. The high-frequency branch splits into two parts: Diff-G computes feature differences between bitemporal high-frequency feature and integrates them into the loss function to improve cross-modal feature

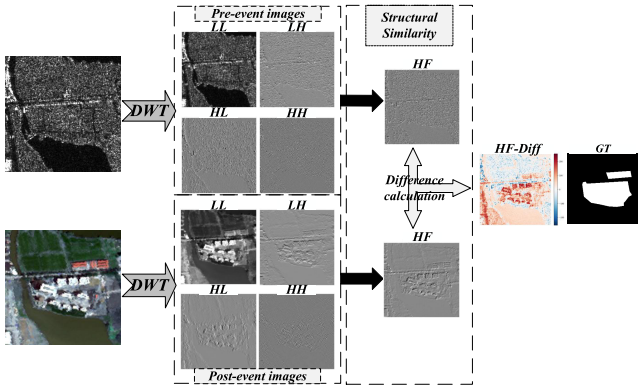


Fig. 1. Illustration of the core principle behind our method. This diagram shows how a discrete wavelet transform (DWT) decomposes heterogeneous images into four frequency subbands: low-frequency (LL), and high-frequency components in horizontal (LH), vertical (HL), and diagonal (HH) directions. By generating a difference map (HF-Diff) primarily from the high-frequency components, which capture shared structural information, our approach effectively mitigates severe spectral discrepancies between the original images. GT denotes the ground truth.

matching, while Dir-G optimizes feature weights based on horizontal and vertical orientations, hence enriching local details and edge information of bitemporal feature. These refined high-frequency features are subsequently merged with low-frequency components through inverse wavelet transform, yielding directionally enhanced spatial-domain features. In the FSD^{2}F module, cross-domain features from different stages are combined to create a frequency–spatial domain difference map that incorporates global structure, high-frequency edges, and fine spatial details. Finally, a lightweight convolution decoder generates the final CD mask.

Our contributions are summarized as follows.

1) We introduce two scalable modules: Diff-G is designed to enhance cross-domain difference extraction by computing high-frequency differences between bitemporal images, and integrating them into the loss function. Dir-G is designed to refine local details and edge information in horizontal and vertical directions using wavelet-based high-frequency features.
2) We propose the FSD^{2}F module to integrate frequency-domain and spatial-domain difference features, replacing conventional feature fusion via single-dimensional concatenation. By combining wavelet-extracted high-frequency components with spatial-domain features, this module strengthens difference extraction and facilitates cross-domain difference feature fusion.
3) We construct a new dataset for Hete-CD, incorporating both optical and SAR images, offering a more challenging and representative benchmark. This dataset serves as a practical resource for training and evaluating supervised Hete-CD methods.

The structure of the article is organized as follows: Section II reviews existing Hete-CD methods, and discusses the application of wavelet transforms in the field of computer vision. Section III presents a comprehensive explanation of the proposed method, accompanied by an in-depth analysis. Section IV describes the experimental setup, including details

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of the dataset, comparative experiments, ablation studies, and parameter analysis. Finally, Section V summarizes the key findings and contributions of this work.

II. Related Work

A. Heterogeneous Remote Sensing Image CD

With advancements in remote sensing technology and the increasing availability of multisensor and multiplatform data, Hete-CD has increasingly attracted academic interest. However, spectral differences in HRSI remain a major challenge. To address this, researchers have explored various strategies for capturing spatial dependencies [26], [27], [28], [29]. These methods not only focus on pixel-level changes but also consider interrelationships, thus enhancing accuracy and robustness. For example, Sun et al. [26] proposed an algorithm based on nonlocal patch similarity, which established relationships between heterogeneous images by constructing a graph structure. Similarly, Chen et al. [29] employed an object-based image analysis approach, generating a structural graph from preprocessed multimodal image pairs and learning vertex and edge information. On the other hand, Touati et al. [15] introduced a Markov model to capture robust visual cues, leveraging their invariance to changes in the image.

In addition to directly computing the structural properties, many researchers [18], [30], [31], [32] have sought to transform heterogeneous image into a homogeneous one to mitigate the spectral differences. For example, Zhan et al. [33] proposed a logarithmic transformation feature learning framework that applies logarithmic transformation to SAR images, enabling them to have similar statistical distribution properties as optical images. Lv et al. [34] utilized a cycle-consistent GAN (CGAN) to convert preevent images into postevent images and incorporated these converted images into their proposed E-UNet network for Hete-CD. Similarly, Wu et al. [18] used a generalized autoencoder to transform heterogeneous image representations into another.

However, such feature transformation methods cannot ensure the quality of the transformed images and may introduce errors. To address this limitation, many approaches project heterogeneous image features into a shared feature space, thus minimizing spectral differences in CD. For example, Liu et al. [35] mapped optical and SAR images into a shared feature space, achieving cross-modal feature reconstruction and reducing spectral differences. Similarly, Ma et al. [36] proposed a Hete-CD method based on dual image translation and contrastive learning, which aligned source and target images more accurately and significantly improved detection accuracy.

Despite these advances, existing methods exhibit notable shortcomings: transformation-based techniques risk degrading image quality and introducing artifacts, while shared feature space approaches often overlook fine-grained edge details critical for accurate CD, particularly in noisy or complex scenes such as SAR imagery. Moreover, many of these methods rely heavily on global feature alignment, which may fail to capture localized, subtle changes effectively.

B. Wavelet Transform-Based Image Processing

In remote sensing, wavelet transforms are widely utilized for tasks such as denoising and classification [37], [38], [39]. For instance, Duan et al. [38] improved SAR image segmentation by convolutional wavelet neural networks and Markov random fields, while Bascoy et al. [39] employed wavelet-based feature extraction for hyperspectral image denoising. Additionally, Yang et al. [37] enhanced classification performance by integrating frequency and spatial information through a wavelet transform feature decomposer (WTFD).

In CD tasks, wavelet transforms have been paired with CNNs and attention mechanisms to bolster noise robustness and improve feature extraction [40], [41]. For example, Wang et al. [40] incorporated wavelet transforms into an encoder–decoder framework with attention modules to enhance detail recovery. Similarly, Xiong et al. [41] proposed a method that fuses frequency-domain features with attention mechanisms to improve CD outcomes. However, these approaches primarily address homogeneous images CD (Homo-CD) and are less explored in Hete-CD. Other frequency-domain CD methods, such as those proposed by Zhou et al. [42] and Chen et al. [27], emphasize detail retention and modal heterogeneity through frequency selection and Fourier domain analysis.

However, these approaches exhibit several key limitations when applied to Hete-CD. First, many prominent wavelet-based methods [41] are designed for Homo-CD. They primarily use frequency features to enhance detail and reduce noise, but do not fundamentally address the large modality gap inherent. Second, other frequency-domain methods [27], [42] often treat frequency information as a supplementary input to be fused with spatial features. They lack a mechanism to explicitly leverage the structural consistency in high-frequency bands to guide the backbone network's feature learning process from end to end. Consequently, these methods often overlook the powerful role that high-frequency similarity can play in enforcing modality-invariance.

To address these limitations, we propose a novel Hete-CD method that uses high-frequency wavelet features not just as another source of information, but as a guiding signal. Our approach leverages the cross-domain similarity and directional characteristics of wavelet features to resolve discrepancies in heterogeneous images. By actively supervising the network to learn modality-invariant high-frequency representations and explicitly modeling directional details, our method significantly enhances CD accuracy.

III. Methodology

To overcome the limitations of detail extraction in heterogeneous image processing and to improve the interpretability of CD across multimodal imagery, we propose WaveHFG, as illustrated in Fig. 2.

The core concept is to exploit the similarity and directionality of heterogeneous images in the wavelet frequency domain, enhancing their feature representation capabilities within neural networks. By incorporating high-frequency information to guide the identification of bitemporal differences, the network optimizes feature extraction and comparison in CD. This wavelet-guided approach, by focusing on structural discrepancies evident in high frequencies, also aims to enhance the interpretability of detected changes. The WaveHFG network comprises three main modules: Diff-G, Dir-G, and FSD^{2}F. The collaborative operation of these modules plays a crucial role in improving CD accuracy in HRSI.

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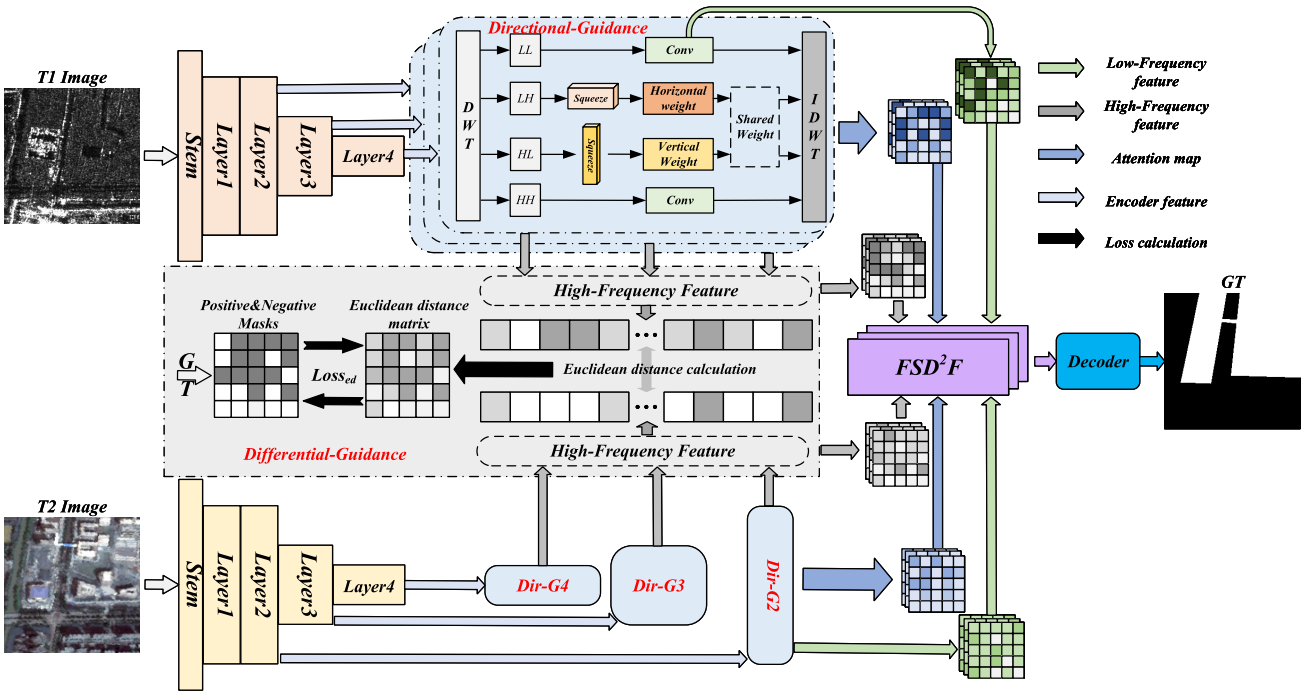


Fig. 2. Construction of a heterogeneous remote sensing image CD network based on WaveHFG.

A. High-Frequency Diff-G

The primary challenge in Hete-CD is guiding the network to learn features consistent across modalities, as traditional spatial comparisons often lead to bias. To address this, the Diff-G module promotes structural consistency via a high-frequency similarity loss (L_{HF}), as shown in Fig. 3. Instead of directly transforming features, this loss calculates the Euclidean distance between bitemporal high-frequency features, which are then adjusted by an adaptive weighting mechanism to handle class imbalance. By encouraging representations to remain similar in unchanged regions and different in changed areas, Diff-G facilitates the extraction of a common structural representation shared by both modalities.

Specifically, the input images are fed into an enhanced ResNet to extract multilevels features. The feature map extracted from the lth layer of DO-Conv [43]-based ResNet34 is denoted as $F_t^l \in \mathbb{R}^{C\times H\times W}$, where $t \in \{A, B\}$. These encoder features are then decomposed into corresponding low-frequency and high-frequency components using a wavelet transform. This process is expressed as follows:

$$\left\{\mathrm{LL}_t^l, \mathrm{LH}_t^l, \mathrm{HL}_t^l, \mathrm{HH}_t^l\right\} = \mathrm{DWT}\left(F_t^l\right), \quad l \in \{1, \ldots, 4\} \tag{1}$$

where DWT represents the discrete wavelet transform, LL denotes the low-frequency component, and LH, HL, and HH denote the high-frequency components. The high-frequency features from each encoder stage l are aggregated using a summation strategy

$$\mathrm{HF}_t^l = \mathrm{LH}_t^l + \mathrm{HL}_t^l + \mathrm{HH}_t^l, l \in \{1, \ldots, 4\}. \tag{2}$$

By aggregating details from horizontal (LH), vertical (HL), and diagonal (HH) orientations, the resulting HF_t^l encapsulates the total structural detail at each pixel location. This summation strategy is motivated by the desire to create a comprehensive feature map that captures edge information from all directions. The main impact is enabling the Diff-G module to focus on the magnitude of structural differences, ignoring their specific orientation.

The label tensor is defined as $L \in \mathbb{R}^{H\times W}$, where each pixel assumes a value of 0 or 1, representing unchanged and changed regions. To facilitate subsequent computations, both the features and labels are flattened into $\mathrm{HF}_t^{\mathrm{flat}} \in \mathbb{R}^{N\times C}$ and $L^{\mathrm{flat}} \in \mathbb{R}^N$, where $N = H \times W$. To mitigate disparities between heterogeneous features, each pixel's C-dimensional high-frequency feature vector is normalized as follows:

$$\widehat{\mathrm{HF}_t^{\mathrm{flat}}} = \frac{\mathrm{HF}_t^{\mathrm{flat}}}{\left\|\mathrm{HF}_t^{\mathrm{flat}}\right\| + \epsilon}, \quad t \in \{A, B\} \tag{3}$$

where $\|\cdot\|$ denotes the $L2$ norm applied to each C-dimensional pixel feature vector in $\mathrm{HF}_t^{\mathrm{flat}}$ individually, and ϵ is a minimal value used to prevent division by zero errors.

Subsequently, the Euclidean distance between the normalized feature vectors is computed as follows:

$$D = \left\|\widehat{\mathrm{HF}_A^{\mathrm{flat}}} - \widehat{\mathrm{HF}_B^{\mathrm{flat}}}\right\| \in \mathbb{R}^N. \tag{4}$$

The masks for positive and negative samples are constructed based on the flattened label tensor L^{flat}

$$\begin{aligned} M_{\mathrm{pos}} &= 1\left(L^{\mathrm{flat}} = 0\right) \in \mathbb{R}^N \\ M_{\mathrm{neg}} &= 1\left(L^{\mathrm{flat}} = 1\right) \in \mathbb{R}^N \end{aligned} \tag{5}$$

where M_{pos} and M_{neg} represent masks created from L^{flat}, $1(\cdot)$ denotes the indicator function, assigning a value of 1 to pixels satisfying the condition and 0 otherwise.

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To mitigate class imbalance between positive and negative samples, dynamic weights are introduced and adaptively adjusted based on their distribution within each batch

$$\begin{aligned} N_{\text{pos}} &= \sum_{i=1}^{N} M_{\text{pos}}[i] + \epsilon \\ N_{\text{neg}} &= \sum_{i=1}^{N} M_{\text{neg}}[i] + \epsilon \end{aligned} \tag{6}$$

$$\begin{aligned} w_{\text{pos}} &= \frac{N_{\text{neg}}}{N_{\text{pos}} + N_{\text{neg}}} \\ w_{\text{neg}} &= \frac{N_{\text{pos}}}{N_{\text{pos}} + N_{\text{neg}}} \end{aligned} \tag{7}$$

where N_{pos} and N_{neg} represent sample counts, ϵ prevents division-by-zero errors, and w_{pos} and w_{neg} denote weights ensuring balanced contributions to the loss function despite varying class proportions.

The loss functions for positive and negative samples are formulated based on the computed distances and weights as follows:

$$\mathcal{L}_{\text{pos}} = w_{\text{pos}} \cdot M_{\text{pos}} \cdot D^2 \tag{8}$$

$$\mathcal{L}_{\text{neg}} = w_{\text{neg}} \cdot M_{\text{neg}} \cdot (\max(1.0 - D, 0))^2 \tag{9}$$

where $\mathcal{L}_{\text{pos}}$ denotes the loss for the positive sample and $\mathcal{L}_{\text{neg}}$ denotes the loss for the negative sample. A Euclidean distance loss function is utilized to increase the separation between the feature vectors of changed regions, using a threshold of 1.0 to delineate changed from unchanged regions.

Finally, the total high-frequency loss is defined as the arithmetic mean of the positive and negative losses

$$\mathcal{L}_{\text{HF}} = \frac{1}{2}\left(\frac{\sum_{i=1}^{N}\mathcal{L}_{\text{pos}}[i]}{N_{\text{pos}}} + \frac{\sum_{i=1}^{N}\mathcal{L}_{\text{neg}}[i]}{N_{\text{neg}}}\right). \tag{10}$$

This balanced combination ensures equitable weighting of the contributions from unchanged and changed regions. Specifically, by separately normalizing the total loss for positive L_{pos} and negative L_{neg} samples by their respective counts N_{pos} and N_{neg}, we calculate the average loss per sample for each class. As a result, the minority class (changed regions) contributes equally to the total loss as the majority class (unchanged regions), preventing the model from being biased toward the more prevalent class.

B. High-Frequency Dir-G

While high-frequency features offer rich details, they are often prone to noise and lack sufficient structural definition. To address this, the Dir-G module refines high-frequency features employs a "cross-mapping" strategy designed to enhance horizontal and vertical components, as shown in Fig. 4. By utilizing an adaptive attention mechanism, this module dynamically adjusts the weights of high-frequency subbands (HL and LH). This effectively enhanced high-frequency details, resulting in more precise boundary localization.

Specifically, a wavelet transform is applied to each layer of the multilayer features, generating four frequency subbands per layer. The resulting subbands, LH and HL, are then processed through adaptive averaging pooling to minimize redundant information, as expressed in the following equation:

$$\begin{aligned} P_{\text{hl}} &= \text{AvgP}(\text{HL}) \in \mathbb{R}^{C\times H\times 1} \\ P_{\text{lh}} &= \text{AvgP}(\text{LH}) \in \mathbb{R}^{C\times 1\times W} \end{aligned} \tag{11}$$

where AvgP denotes average pooling, which compresses the spatial dimensions of HL into a height-only representation and LH into a width-only representation, preserving channel information. Subsequently, these pooled feature are further refined by a shared 1×1 convolutional layer that compresses the channel dimension, followed by batch normalization (BN) and a Swish activation S function to introduce nonlinearity. This process yields

$$\begin{aligned} Y_{\text{hl}} &= S\left(\text{BN}\left(\text{Conv}_{1\times 1}(P_{\text{hl}})\right)\right) \in \mathbb{R}^{C_{\text{mid}}\times H\times 1} \\ Y_{\text{lh}} &= S\left(\text{BN}\left(\text{Conv}_{1\times 1}(P_{\text{lh}})\right)\right) \in \mathbb{R}^{C_{\text{mid}}\times 1\times W}. \end{aligned} \tag{12}$$

These refined features are then used to compute attention weights for HL and LH subbands through independent 1×1 convolutional layers, with the outputs normalized by the Sigmoid activation function

$$\begin{aligned} A_{\text{hl}} &= \sigma(\text{Conv}_h(Y_{\text{hl}})) \in \mathbb{R}^{C\times H\times 1} \\ A_{\text{lh}} &= \sigma(\text{Conv}_w(Y_{\text{lh}})) \in \mathbb{R}^{C\times 1\times W} \end{aligned} \tag{13}$$

where σ denotes the Sigmoid function, and A_{hl} and A_{lh} represent the attention weights for the HL and LH subbands, respectively. A cross-mapping mechanism then enhances directional detail by applying A_{hl} to LH and A_{lh} to HL, integrating horizontal and vertical information

$$\begin{aligned} \text{HL}_{\text{out}} &= \text{HL} \otimes A_{\text{lh}} + \text{HL} \\ \text{LH}_{\text{out}} &= \text{LH} \otimes A_{\text{hl}} + \text{LH} \end{aligned} \tag{14}$$

where $\odot$ indicates element-wise multiplication. This operation amplifies the model's ability to capture both horizontal and vertical details, yielding a more comprehensive representation of changed regions. The cross-mapping operation is formally defined as follows:

$$\{\text{HL}_{\text{out}}, \text{LH}_{\text{out}}\} = \text{WaveX}\{\text{HL}, \text{LH}\}. \tag{15}$$

Meanwhile, the unprocessed HH and LL subbands are processed using the same convolutional block to suppress noise interference and filter out irrelevant variations. Finally, the low-frequency and high-frequency components are reconstructed into spatial-domain features through an inverse wavelet transform

$$\begin{aligned} F^{l}_{\text{dir}_t} &= \text{IDWT}\left\{\text{LL}'_{t_l}, \text{LH}^{l}_{\text{out}_t}, \text{HL}^{l}_{\text{out}_t}, \text{HH}'_{t_l}\right\} \\ & l \in \{1,\ldots,4\}, t \in \{A, B\}. \end{aligned} \tag{16}$$

Through the integration of high-frequency Dir-G and cross-mapping, Dir-G module effectively captures directional details, offering a robust and accurate solution for CD in HRSI.

C. Frequency-Space Domain Difference Fusion

Conventional fusion methods often use channel concatenation. However, this approach fails to fully exploit the complementary information between spatial and frequency domains. To address this, the FSD^{2}F module uses 3-D convolution to model cross-domain interactions, as shown in Fig. 5.

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We stack difference maps from different domains to form a 3-D tensor. Then, a 3-D convolution layer extracts joint patterns from this structure. This method integrates difference features more effectively than standard 2-D fusion.

The primary function of this module is to process both the low-frequency and high-frequency subband features obtained from the wavelet transform in the Dir-G module, along with the subsequently integrated spatial-domain features. The six input feature sets are denoted as $F_{\mathrm{ll}_A}^{l-1}$, $F_{\mathrm{ll}_B}^{l-1}$, HF_A^{l-1}, HF_B^{l-1}, $F_{\mathrm{dir}_A}^{l}$, $F_{\mathrm{dir}_B}^{l}$. Notably, wavelet transform typically reduces the feature dimensions by half. To effectively handle the multilevel features, the low-frequency and high-frequency subband features are extracted from the lower encoder levels, while the spatial-domain features are derived from the higher encoder levels.

The process begins by concatenating each pair of input features to form domain-specific difference features: F_{low}^{l}, F_{high}^{l}, F_{spa}^{l}, representing the low-frequency, high-frequency, and spatial domains, respectively. This step integrates features within each domain, producing a comprehensive set of difference features. To ensure independent processing and maximize the extraction of latent information, separate 2-D convolutional layers are applied, followed by BN and ReLU activation functions. This approach preserves the integrity of each feature domain while maximizing the retention of its unique information. The process is formulated as follows:

$$\begin{aligned} F_{\mathrm{low}}^{l} &= \mathrm{ReLU}\left(\mathrm{BN}\left(\mathrm{Conv}\left(F_{\mathrm{ll}_A}^{l-1} \parallel F_{\mathrm{ll}_B}^{l-1}\right)\right)\right) \\ F_{\mathrm{high}}^{l} &= \mathrm{ReLU}\left(\mathrm{BN}\left(\mathrm{Conv}\left(\mathrm{HF}_A^{l-1} \parallel \mathrm{HF}_B^{l-1}\right)\right)\right) \\ F_{\mathrm{spa}}^{l} &= \mathrm{ReLU}\left(\mathrm{BN}\left(\mathrm{Conv}\left(F_{\mathrm{dir}_A}^{l} \parallel F_{\mathrm{dir}_B}^{l}\right)\right)\right) \end{aligned} \tag{17}$$

where $\parallel$ denotes the concatenation operation.

For frequency-spatial feature fusion, the three processed 2-D feature sets are stacked along the depth dimension to form a 3-D feature tensor. A 3-D convolutional layer with a kernel size and stride of 3 is then employed to capture cross-domain relationships, facilitating effective fusion and integration of complementary information. The output channels of this layer are denoted as C_{out}, with the feature dimensions is $H \times W$. This operation is expressed as follows:

$$\begin{aligned} F_{\mathrm{stack}}^{l} &= \mathrm{Stack}\left(F_{\mathrm{low}}^{l}, F_{\mathrm{high}}^{l}, F_{\mathrm{spa}}^{l}\right) \in \mathbb{R}^{C_{\mathrm{out}} \times 3 \times H \times W} \\ F_{\mathrm{fusion}}^{l} &= \mathrm{ReLU}\left(\mathrm{BN}\left(\mathrm{Conv3D}\left(F_{\mathrm{stack}}^{l}\right)\right)\right). \end{aligned} \tag{18}$$

This operation not only fuses spatial information effectively but also exploits the complementary nature of the feature maps along the depth dimension. To further enhance feature integration, the module incorporates residual connections. These connections preserve feature information throughout the learning process, mitigating risks such as vanishing or exploding gradients, and ensuring robust cross-dimensional fusion.

D. Backbone for Guided Hierarchical Feature Extraction

The WaveHFG architecture employs a backbone encoder, such as our current enhanced ResNet34 with DO-Conv [43], to extract hierarchical multilevel features $F_t^l \in \mathbb{R}^{C \times H \times W}$ ($t \in \{A, B\}$, l denotes layer) from heterogeneous image pairs. This backbone is architecturally modular, allowing substitution with other advanced encoders based on application or resource needs. Its learnable parameters are dynamically modulated via our wavelet-domain guidance.

This guidance strategy mitigates modality-specific biases and fosters sensor-invariant representations. Backbone-extracted features F_t^l are inputs for wavelet decomposition. Crucially, the similarity loss $\mathcal{L}_{\mathrm{HF}}$ [see (10)], derived from Euclidean distances between wavelet high-frequency representations of F_A^l and F_B^l as detailed in the Diff-G module (Section III-B), is backpropagated through the entire network. This directly influences backbone weight updates, compelling it to learn spatially rich features with superior cross-modal comparability in their high-frequency structural content.

Optimizing the backbone for robust wavelet frequency-domain feature alignment via this high-frequency similarity guidance effectively curtails modality bias. This facilitates extracting shared, sensor-invariant characteristics pivotal for robust Hete-CD, as the backbone's learnable filters progressively adapt to these consistency objectives.

E. Loss Function

Cross-entropy (CE) loss is employed to measure the discrepancy between the predicted change region and the ground truth labels. For each pixel, the model outputs probabilities for "change" and "no change," with the true label assigned as either 0 or 1. The CE loss is calculated as follows:

$$\mathcal{L}_{\mathrm{CE}} = -\frac{1}{N}\sum_{i=1}^{N}\left(y_i \log\left(p_{\mathrm{c},i}\right) + (1 - y_i)\log\left(p_{\mathrm{noc},i}\right)\right) \tag{19}$$

where N represents the total number of pixels, y_i denotes the true label for pixel i, and $p_{\mathrm{c},i}$ and $p_{\mathrm{noc},i}$ are the predicted probabilities of "change" and "no change."

In addition to the primary classification loss $\mathcal{L}_{\mathrm{CE}}$, we incorporate our high frequency loss, denoted as $\mathcal{L}_{\mathrm{HF}}$ [defined in (10)]. This $\mathcal{L}_{\mathrm{HF}}$ term encourages the backbone network to learn modality-invariant structural features by quantifying consistency in wavelet high-frequency representations. The total loss function, $\mathcal{L}_{\mathrm{Total}}$, is a weighted sum of these two components

$$\mathcal{L}_{\mathrm{Total}} = \mathcal{L}_{\mathrm{CE}} + \alpha \cdot \mathcal{L}_{\mathrm{HF}} \tag{20}$$

where α is a hyperparameter that balances the contribution of the high-frequency guidance. The selection of α is based on empirical validation, as discussed in our parametric analysis. This composite loss thereby guides the model to optimize for both accurate pixel-wise CD and robust cross-modal feature extraction.

IV. Experimental Results and Analysis

A. Implementation Details

In this study, all experiments are conducted on two NVIDIA GeForce RTX 4090 GPUs running Ubuntu 20.04.6 LTS. The implementation is based on PyTorch 2.4.1 with CUDA 12.1 and Python 3.10. We use the AdamW optimizer [44] with an initial learning rate of 0.0001, a weight decay of 0.00005, a batch size of 16, and train for 200 epochs. To mitigate overfitting and enhance generalization, standard data augmentation strategies including random cropping, rotation, and flipping are applied. After each epoch, the model is evaluated on the

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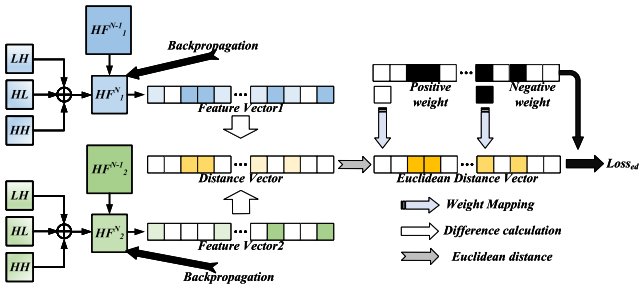


Fig. 3. Structure of the high-frequency Diff-G.

validation set, and the checkpoint with the best validation performance is retained for testing. To reduce randomness and ensure fairness, all experiments are repeated five times with different random seeds, and the reported results are averaged over these independent runs.

During the testing phase, five evaluation metrics are employed to assess the model performance: Precision (Pre), Recall (Re), $F1$-score ($F1$), intersection over union (IoU), and overall accuracy (OA). Among these, the $F1$-score and IoU serve as the primary metrics, with higher values indicating superior performance. The definitions of these metrics are as follows:

$$\begin{aligned} \text{Pre} &= \frac{\text{TP}}{\text{TP}+\text{FP}} \\ \text{Re} &= \frac{\text{TP}}{\text{TP}+\text{FN}} \\ \text{F1} &= \frac{2\cdot\text{Pre}\cdot\text{Re}}{\text{Pre}+\text{Re}} \\ \text{IoU} &= \frac{\text{TP}}{\text{TP}+\text{FN}+\text{FP}} \\ \text{OA} &= \frac{\text{TP}+\text{TN}}{\text{TP}+\text{TN}+\text{FP}+\text{FN}} \end{aligned} \tag{21}$$

where TP, TN, FP, and FN denote the numbers of true positives, true negatives, false positives, and false negatives, respectively.

B. Datasets

The U.K.-HCD dataset [45] was captured in the Gloucester region of the U.K., spanning July 2006–July 2007. It is primarily designed to detect land cover changes following flooding events. The dataset comprises remote sensing images with a spatial resolution of 0.65 m/pixel. The initial image was acquired from the Pleiades satellite, while the subsequent image was obtained from the TerraSAR-X satellite. The images have dimensions of 2325 × 4135 pixels. For experimentation, the images were divided into 256 × 256 pixel blocks, yielding 144 samples: 31 target samples and 113 nontarget samples, the latter characterized by a uniformly black appearance. The dataset was randomly partitioned into three subsets: 117 samples for training, 14 for validation, and 14 for testing.

The France-HCD dataset [45] was captured in the Toulouse region of France, spanning from February 2009 to July 2013. It is primarily designed to detect land changes during the construction process. The dataset comprises remote sensing images with a spatial resolution of 2 m/pixel. The initial image was acquired from the Pleiades satellite, while the subsequent image was obtained from the TerraSAR-X satellite. The images have dimensions of 4404 × 2604 pixels. For experimental purposes, the images were cropped into 256 × 256 pixel blocks, yielding a total of 170 samples, including 68 target samples and 102 nontarget samples. The dataset is divided into three subsets: 111 samples for training, 32 samples for validation, and 32 samples for testing.

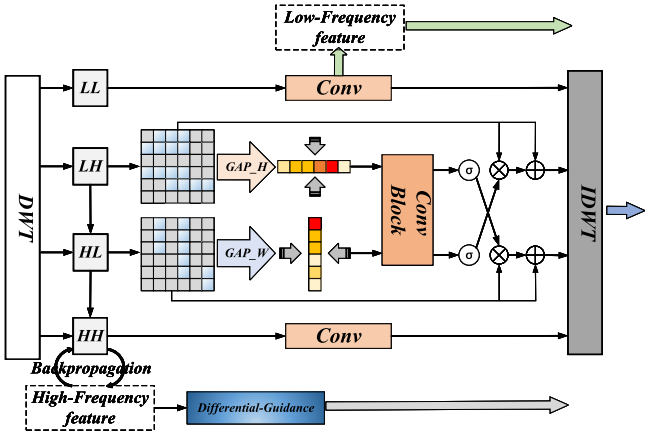


Fig. 4. Structure of the high-frequency Dir-G.

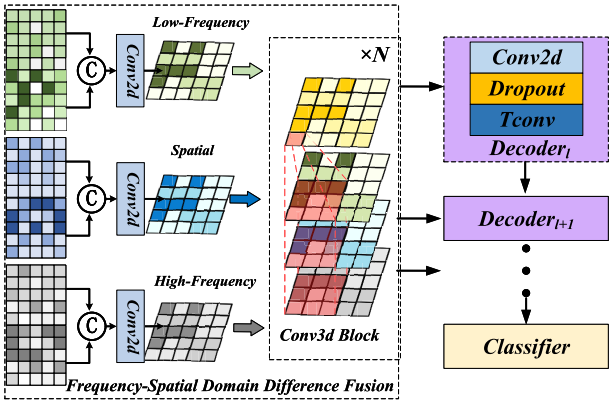


Fig. 5. Structure of the frequency-space difference fusion.

The GF-HCD is a newly constructed dataset captured in the border region between Shanghai, Suzhou, and Jiaxing, China, spanning 2019–2024, as shown in Fig 6. It focuses on land cover, building, and road changes, reflecting the urbanization process and land use transformation over time. The dataset consists of two sets of satellite images obtained from different remote sensing platforms. The first set includes optical images captured by the Gaofen-7 satellite, providing detailed surface feature information, including buildings, roads, vegetation, and water bodies. The second set includes SAR images from the Gaofen-3 satellite, offering all-weather, continuous observation of ground changes under varying meteorological conditions. The dataset includes two images with dimensions of 5662 × 10 259 pixels and 2913 × 11 520 pixels, respectively. After cropping into 256 × 256 pixel blocks, the dataset contains 1375 samples, including 666 target samples and 709 nontarget samples. The dataset is divided into three subsets: 1123 samples for training, 130 samples for validation, and 130 samples for testing.

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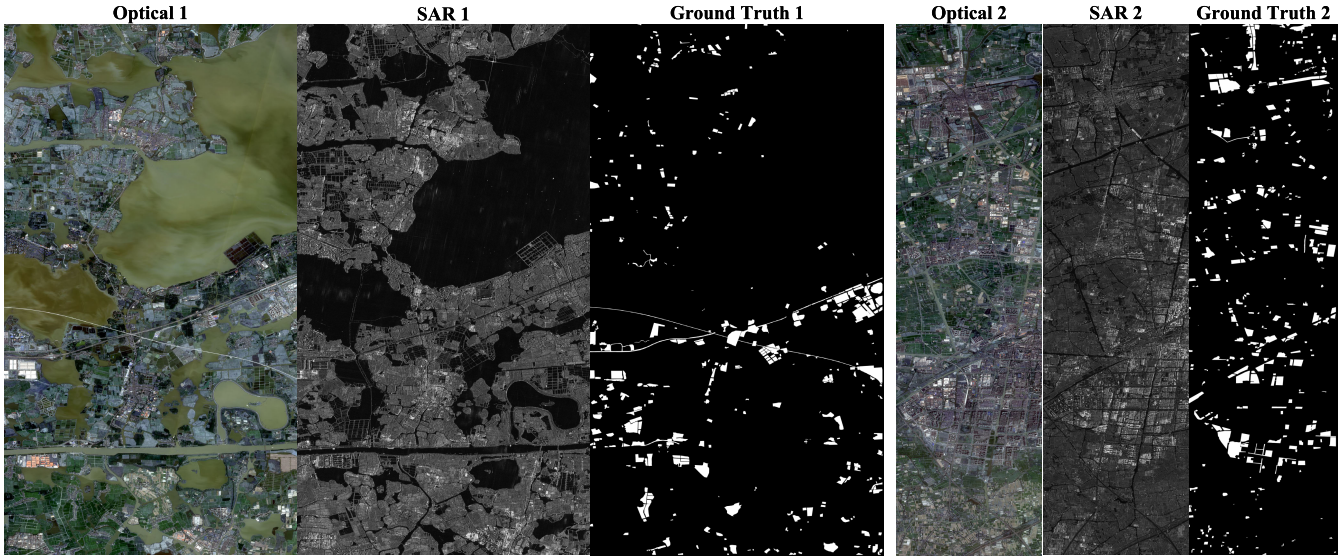


Fig. 6. Visualization of the proposed GF-HCD dataset. All samples are shown, each consisting of a preevent optical image (first column), a postevent SAR image (second column), and the corresponding ground truth map (third column). In the ground truth maps, white pixels indicate changed regions, and black pixels indicate unchanged regions.

TABLE I

IMPACT OF DIFFERENT FACTORS OF HIGH-FREQUENCY SIMILARITY LOSS ON DATASET PERFORMANCE

Factors	UK-HCD		France-HCD		GF-HCD	
	F1	IoU	F1	IoU	F1	IoU
0.1	88.72	79.73	53.79	37.34	62.36	45.37
0.15	**90.06**	**81.92**	**59.07**	**41.95**	65.43	48.63
0.2	87.66	78.04	57.14	40.15	64.98	48.13
0.25	88.48	79.36	53.99	37.04	**66.75**	**50.10**
0.3	88.60	79.56	55.49	38.42	62.81	45.89

TABLE II

ABLATION STUDY RESULTS ON THREE DATASETS (U.K.-HCD, FRANCE-HCD, GF-HCD)

Module			UK-HCD		France-HCD		GF-HCD	
Dir-G	Diff-G	FSD^{2}F	F1	IoU	F1	IoU	F1	IoU
×	×	×	85.56	74.77	54.76	37.71	60.04	42.97
✓	×	×	86.50	76.23	56.73	39.68	63.91	46.96
×	✓	×	86.47	76.17	56.89	39.85	61.15	44.05
×	×	✓	85.72	75.01	56.91	39.78	63.76	46.82
✓	✓	×	86.56	76.36	57.22	40.15	64.19	47.27
✓	×	✓	87.11	77.17	58.05	40.90	64.02	47.11
×	✓	✓	86.88	76.83	57.26	40.13	65.05	48.21
✓	✓	✓	**90.06**	**81.92**	**59.08**	**41.95**	**66.75**	**50.10**

C. Parametric Analysis

To investigate the influence of weighting factors for $\mathcal{L}_{\text{HF}}$ on model performance, we conducted a parameter analysis, with results summarized in Table I. Variations in these factors yielded noticeable changes in both $F1$-scores and IoU values across the U.K.-HCD, France-HCD, and GF-HCD datasets.

With weighting factors set to 0.1, the model exhibited relatively low performance across all three datasets, with $F1$-scores of 88.72%, 53.79%, and 62.36%, and IoU values of 79.73%, 37.34%, and 45.37% on U.K.-HCD, France-HCD, and GF-HCD, respectively. Increasing the weighting factor to 0.15 led to the best results on U.K.-HCD (90.06%/81.92%) and France-HCD (59.07%/41.95%), suggesting that moderate high-frequency constraints are most suitable for smaller datasets with limited diversity.

In contrast, further increases to 0.20 caused a decline on France-HCD (57.14%/40.15) and a slight decrease on U.K.-HCD (87.66%/78.04), but GF-HCD maintained competitive performance (64.98%/48.13). At 0.25, GF-HCD achieved its peak with 66.75% $F1$ and 50.10% IoU, while U.K.-HCD remained stable (88.48%/79.36). This indicates that larger datasets can accommodate stronger high-frequency constraints, which may help capture more subtle variations without severe overfitting.

When the weighting factor reached 0.30, performance on France-HCD and GF-HCD started to decline (55.49%/38.42 and 62.81%/45.89, respectively), reflecting the risk of overemphasizing noise or minor details. By contrast, U.K.-HCD remained relatively stable (88.60%/79.56), though without further gains beyond its earlier peak.

Overall, smaller datasets such as U.K.-HCD and France-HCD exhibit optimal performance at moderate weighting (0.15), but are more sensitive to further increases. In contrast, the larger GF-HCD dataset benefits from a slightly higher weighting (0.25), underscoring the role of dataset scale in determining the optimal balance of high-frequency supervision. The observed U-shaped performance trend suggests that excessively low weights provide insufficient structural guidance, while overly high weights emphasize noise, with the optimal point shifting upward as dataset size increases.

D. Ablation Experiment

In this section, we conduct a series of ablation experiments to validate the effectiveness of our three core modules: Dir-G for directional enhancement, Diff-G for Diff-G, and FSD^{2}F for cross-domain fusion. The results are presented in Table II.

Our baseline model, corresponding to the first row of Table II, is defined as a simple encoder-decoder architecture. It utilizes the DO-Conv-based ResNet-34 backbone for feature extraction and a standard convolutional decoder. This

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baseline intentionally omits all of our proposed modules (Dir-G, Diff-G, and FSD^{2}F) and the high-frequency loss L_{HF}. The difference between bitemporal features is computed via subtraction. Each subsequent row in Table II evaluates the contribution of our modules by integrating them into this baseline.

The baseline model exhibited limited performance. It achieved $F1$-scores of 85.56% on U.K.-HCD, 54.76% on France-HCD, and 60.04% on GF-HCD. These results indicate that the baseline model struggles to effectively handle heterogeneous data.

The Dir-G module focuses on differential feature extraction. Applied individually, it improved the $F1$-score by 2.13% on France-HCD and 1.11% on GF-HCD. This demonstrates its capability in detecting subtle yet critical changes.

The Diff-G module, which focuses on differential feature extraction from bitemporal images, also yielded significant gains. Applied individually, it improved the $F1$-score to 56.89% on France-HCD and 61.15% on GF-HCD, demonstrating its ability to detect subtle yet critical changes.

The FSD^{2}F module fuses frequency- and spatial-domain features. When used alone, it raised the $F1$-score by 3.72% on GF-HCD and 2.15% on France-HCD. These gains confirm its effectiveness in complex environments where heterogeneous information must be integrated.

Integrating multiple modules further improved performance across all datasets. The combined application of Dir-G and Diff-G reached an $F1$-score of 86.56% on U.K.-HCD and 64.19% on GF-HCD. This improvement occurs because Diff-G guides the encoder to learn high-frequency features. Meanwhile, Dir-G acts as a fine-grained refiner, sharpening directional details to define change boundaries. Similarly, integrating Diff-G and FSD^{2}F increased scores consistently, notably raising the GF-HCD $F1$-score to 65.05% and the France-HCD score to 57.26%. This demonstrates that the FSD^{2}F fusion mechanism works effectively with the high-quality inputs provided by Diff-G. Furthermore, pairing Dir-G and FSD^{2}F improved the $F1$-score to 87.11% on U.K.-HCD and 58.05% on France-HCD. This indicates that the fusion process utilizes the precise boundary information supplied by Dir-G.

The best performance was achieved when all three modules were integrated. This yielded $F1$-scores of 90.06%, 59.08%, and 66.75% on U.K.-HCD, France-HCD, and GF-HCD, respectively. Compared to the baseline, the full model achieved substantial improvements of 4.50%, 4.32%, and 6.71% on the respective datasets. This demonstrates the strong cooperation among Dir-G, Diff-G, and FSD^{2}F, effectively addressing diverse scenarios and improving robustness across different scales.

In conclusion, the ablation experiments confirm the critical role of each module in enhancing model performance. When integrated, they substantially improve both $F1$-score and IoU values. To minimize randomness and ensure reliability, all reported ablation results are averaged over five independent runs.

E. Comparative Experiment

1) Comparison Method: In our comparative experiments, we benchmarked our proposed method against a suite of advanced CD techniques, encompassing both Homo-CD and Hete-CD approaches, with a primary focus on leading CNN and Transformer-based architectures. Acknowledging that a significant portion of current Hete-CD research gravitates toward unsupervised methodologies, our choice of supervised Hete-CD baselines is consequently focused, primarily drawing from state-of-the-art, publicly available, and open-source works from the past three years. All compared methods were implemented using their publicly available GitHub code and evaluated under consistent experimental parameters. The selected state-of-the-art Homo-CD methods include DTCDSCN [46], SNUNet [47], BIT [48], ChangeFormer [49], DMINet [50], AMTNet [51], AERNet [52], SCanNet [53], MDIPNet [8], and UAB-CD [54]. For Hete-CD, we compared against prominent methods: HAFF [20], AFENet [55], and HFA-PANet [56].

2) Experimental Data and Visualization Results: The effectiveness of our approach was evaluated using the U.K.-HCD and France-HCD datasets, with comprehensive performance metrics reported in Table III. Performance across models on the U.K.-HCD dataset exhibits considerable variation. Models prioritizing precision, such as DTCDSCN, excel in accurately predicting change regions, achieving a precision of 99.21%. However, this comes at the expense of recall, which drops to 53.65%, suggesting that an emphasis on precision may fail to capture the full extent of changed areas. In contrast, models favoring recall, like AFENet (86.63%) and ChangeFormer (85.84%), excel at detecting a wider array of change regions, though often at the cost of reduced precision due to difficulties in distinguishing true changes from false positives, thereby elevating false detection rates. The $F1$-score, which balances precision and recall, provides a more comprehensive evaluation by harmonizing precision and recall. Alongside our proposed WaveHFG method, the best performing methods in $F1$-score include AFENet (89.80%) and ChangeFormer (87.37%), both of which achieve a robust equilibrium between precision and recall. Notably, WaveHFG outperforms all other models in terms of $F1$-score and IoU. For instance, WaveHFG achieves 90.06% $F1$ and 81.92% IoU on U.K.-HCD, and 59.08% $F1$ and 41.95% IoU on France-HCD, demonstrating its ability to effectively distinguish between changed and unchanged regions while maintaining high accuracy. Additionally, on the U.K.-HCD dataset, WaveHFG achieves the highest OA of 98.35%, highlighting its robust adaptability to various change types, particularly in complex disaster-related scenarios. On the France-HCD dataset, WaveHFG attains a strong OA of 94.14%, which is slightly lower than SNUNet (94.33%), indicating that lightweight dense architectures can remain competitive under certain conditions.

The lower performance of BIT can be attributed to its shallower encoder structure and reliance on Transformer-based differential feature generation. In heterogeneous images with significant variations, this limited feature extraction capability adversely affects performance. Similarly, AERNet's lower metrics may result from overfitting; despite strong validation set performance, its test set results decline markedly. HAFF's suboptimal performance is likely caused by its complex architecture and memory constraints, which restrict the batch size to just 3, thus hindering effective feature learning.

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TABLE III

COMPARISON OF MODELS ON U.K.-HCD AND FRANCE-HCD DATASETS. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN RED AND BLUE, RESPECTIVELY

Model	UK-HCD					France-HCD				
	Pre	Recall	F1	IoU	OA	Pre	Recall	F1	IoU	OA
DTCDSCN (GRSL 2021)	**99.21**	53.65	69.60	53.42	96.07	61.25	32.16	42.17	26.72	92.85
SNUNet (GRSL 2021)	**97.45**	57.99	72.71	57.12	96.34	80.71	39.51	**53.05**	**36.10**	**94.33**
BIT (TGRS 2021)	78.67	29.62	43.04	27.42	93.41	71.49	30.78	43.03	27.42	93.40
ChangeFormer (IGARSS 2022)	88.95	85.84	87.37	77.57	97.91	51.18	30.01	37.83	23.33	92.01
DMINet (TGRS 2023)	88.05	50.50	64.19	47.26	95.26	**84.81**	23.72	37.07	22.76	93.47
AMTNet (ISPRS 2023)	88.00	73.40	80.04	66.72	96.92	64.83	27.20	38.32	23.70	92.90
AERNet (TGRS 2023)	96.90	52.49	68.09	51.62	95.86	75.05	34.25	47.03	30.75	93.75
SCanNet (TGRS 2024)	95.55	77.15	85.37	74.47	97.78	59.28	**43.25**	50.01	33.35	92.99
MDIPNet (TIP 2024)	86.31	81.50	83.84	72.17	97.36	71.81	34.77	46.86	30.60	93.61
UAB-CD (ISPRS 2025)	87.74	69.52	77.58	63.37	96.62	55.88	39.86	46.53	30.32	92.58
HAFF (TGRS 2023)	89.62	65.69	75.81	61.05	96.47	68.73	29.66	41.43	26.13	93.21
AFENet (TGRS 2024)	93.20	**86.63**	**89.80**	**81.48**	**98.34**	76.89	33.10	46.28	30.11	93.77
HFA-PANet (PR 2025)	91.04	83.44	87.07	77.11	97.92	**84.71**	33.21	47.61	31.24	**94.08**
WaveHFG	91.62	**88.65**	**90.06**	**81.92**	**98.35**	71.26	**52.68**	**59.08**	**41.95**	94.14

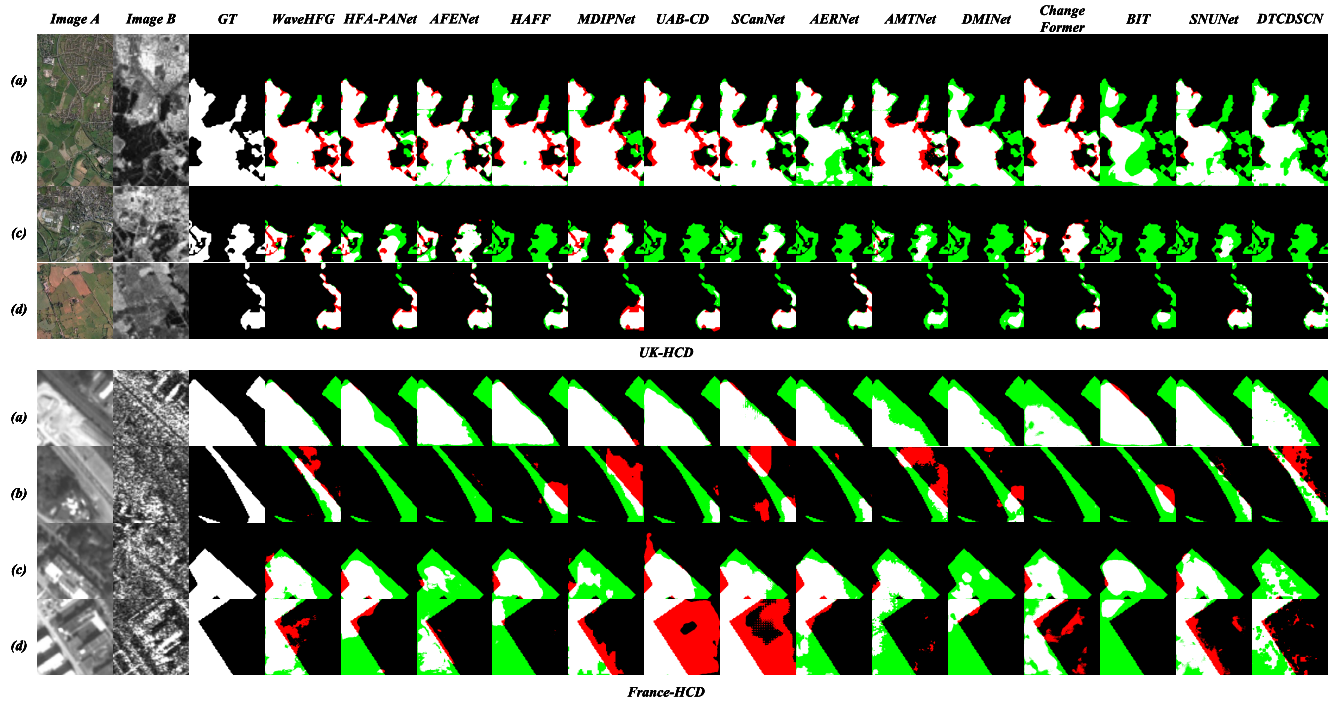


Fig. 7. Visualization results of each method on the U.K.-HCD and France-HCD datasets. Rows (a)–(d) in each dataset represent four different representative change samples. In the result maps, different colors indicate different detectioroutcomes: white represents true positives, black represents true negatives, red represents false positives, and greenrepresents false negatives.

Experiments on the France-HCD dataset reveal that SCanNet and SNUNet exhibit strong performance across multiple metrics, notably $F1$-score and IoU. Unlike disaster assessment imagery with well-defined boundaries, the SAR images in this dataset are characterized by intricate details. These properties, particularly in capturing building structures, introduce variability in feature detection due to light reflections. Consequently, many deep feature extraction methods may over-extract features, obscuring the identification of common patterns. For instance, while ChangeFormer performs well on the U.K.-HCD dataset, its performance on the France-HCD dataset declines significantly. In contrast, SNUNet's simpler dense network architecture achieves superior performance on this dataset, underscoring its adaptability to the complexities of SAR imagery.

The visualization results comparing different methods on the U.K.-HCD and France-HCD datasets are presented in Fig. 7. In the figure, white represents true positives, black represents true negatives, red represents false positives, and green represents false negatives. The results highlight the superior generalization ability of WaveHFG. Methods such as UAB-CD, AMTNet, and SCanNet perform well in Fig. 7(a) and (b) but reveal substantial under-identification in (c) and (d), as evidenced by prominent green regions. This suggests that, despite their precision in detecting changes under certain conditions, these methods falter in more complex

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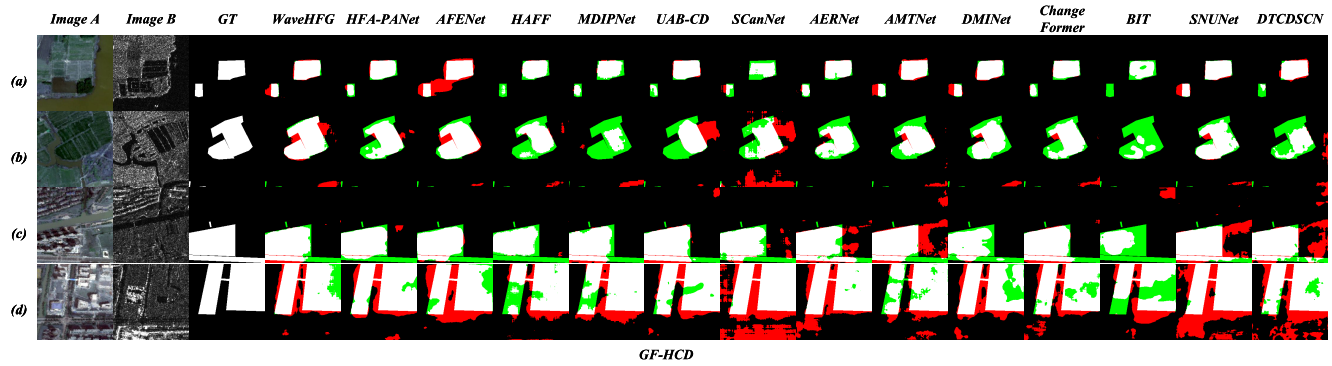


Fig. 8. Visualization results of each method on the GF-HCD datasets.

TABLE IV
COMPARISON OF MODELS ON GF-HCD DATASET. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN RED AND BLUE, RESPECTIVELY

Model	GF-HCD				
	Pre	Recall	F1	IoU	OA
DTCDSCN (GRSL 2021)	57.93	48.08	52.55	35.64	95.69
SNUNet (GRSL 2021)	68.29	56.90	62.08	45.01	96.55
BIT (TGRS 2021)	74.56	31.48	44.27	28.43	96.07
ChangeFormer (IGARSS 2022)	76.14	56.65	64.96	48.11	96.97
DMINet (TGRS 2023)	64.13	53.48	58.32	41.17	96.21
AMTNet (ISPRS 2023)	67.70	58.02	62.49	45.44	96.54
AERNet (TGRS 2023)	64.26	54.81	59.16	42.01	96.24
SCanNet (TGRS 2024)	61.53	46.69	53.09	36.14	95.90
MDIPNet (TIP 2024)	73.73	40.20	52.03	35.16	96.32
UAB-CD (ISPRS 2025)	**79.00**	56.34	65.77	49.00	**97.09**
HAFF (TGRS 2023)	76.18	36.92	49.74	33.10	96.30
AFENet (TGRS 2024)	70.90	**59.78**	64.86	48.00	96.78
HFA-PANet (PR 2025)	**80.28**	56.11	**66.05**	**49.43**	**97.11**
WaveHFG (Ours)	73.91	**61.09**	**66.75**	**50.10**	96.87

scenarios. ChangeFormer performs effectively in Fig. 7(a)–(c), but also shows areas of neglect in (d), indicating its potential inability to capture all alterations in specific cases. In contrast, WaveHFG demonstrates consistent performance across all image data. While its performance in detail recognition may be slightly lower than some other methods, its OA and generalization ability significantly outperform the alternatives. This stability demonstrates WaveHFG's capacity to maintain reliable performance across diverse images and change scenarios, even with limited data.

The GF-HCD dataset, characterized by its larger scale compared to U.K.-HCD and France-HCD, provides a more comprehensive benchmark for evaluating model generalization. The results are presented in Table IV. With the increased data volume, most methods achieve more stable performance, mitigating the overfitting tendencies observed on smaller datasets. For instance, HFA-PANet achieves an $F1$-score of 66.05% and an IoU of 49.43%, ranking second only to our proposed WaveHFG. WaveHFG attains the highest Recall (61.09%), $F1$-score (66.75%), and IoU (50.10%), confirming its robustness in capturing subtle changes and maintaining structural consistency across heterogeneous inputs. Although HFA-PANet and UAB-CD achieve slightly higher OA (97.11% and 97.09%, respectively) than WaveHFG (96.87%), this primarily reflects their emphasis on precision. In contrast, WaveHFG provides a more balanced solution by improving recall without sacrificing overall detection quality, which is particularly important in real-world Hete-CD scenarios.

The visualization results of the comparison methods on the GF-HCD dataset are shown in Fig. 8. As illustrated, WaveHFG excels at synthesizing and accurately identifying key components across all images. For example, in Fig. 8(b), many methods struggle to fully recognize the change target, with only WaveHFG and SNUNet successfully capturing it with higher precision. However, in Fig. 8(c) and (d), SNUNet tends to generate more false positives (highlighted in red), while WaveHFG maintains superior accuracy in these regions. This consistency underscores WaveHFG's robustness in addressing complex scenarios effectively.

A performance analysis across Tables III and IV reveals a key tradeoff. Some methods, like DTCDSCN, are very conservative. They achieve high precision but miss subtle changes, resulting in low recall. This makes them less effective for complete change assessment.

In contrast, our WaveHFG method consistently achieves a state-of-the-art recall rate. This is due to its high-frequency guidance design. The network becomes highly sensitive to all structural differences and captures even the smallest details. The downside is that strong noise, like SAR speckle, is sometimes misclassified as change. This increases false positives and lowers precision. However, this tradeoff is beneficial. WaveHFG achieves the best overall $F1$-score, showing a superior balance between precision and recall. High recall is critical in applications like disaster monitoring, making WaveHFG a more effective solution.

To clearly demonstrate the performance of each method across different datasets, a bar chart summarizing the $F1$-scores for the three datasets is presented in Fig. 9. As depicted, performance varies significantly across methods. For example, Transformer-based methods such as ChangeFormer and AMTNet exhibit substantial declines on the France-HCD dataset. This suggests that, while Transformer architectures excel in certain contexts, their effectiveness can be highly sensitive to dataset characteristics. WaveHFG's primary strength lies in its novel integration of wavelet frequency domain features with high-frequency guidance. Conventional CD methods often focus on global features or basic image differentiation techniques, frequently overlooking the critical role of high-frequency information in defining change boundaries. In

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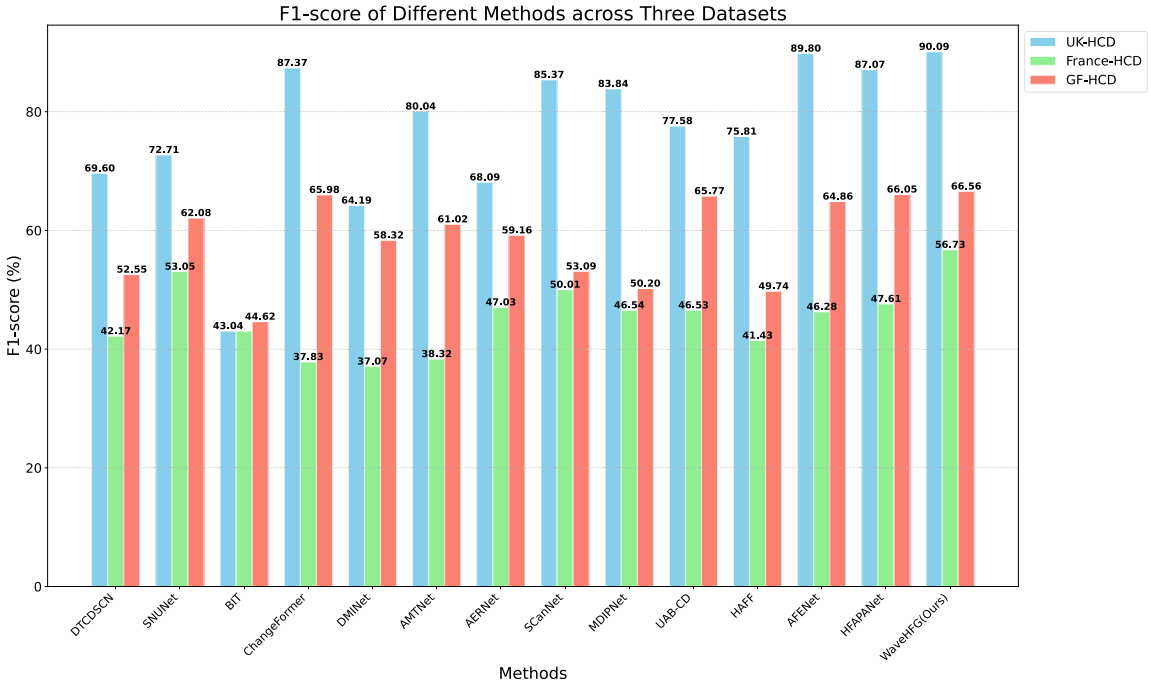


Fig. 9. $F1$-score of SOTA methods on three datasets.

contrast, WaveHFG enhances the detection of subtle changes by leveraging high-frequency components within the wavelet frequency domain. This approach is particularly effective for capturing fine details and addressing complex backgrounds.

Comparative experiments revealed that overly complex networks, particularly those relying on deep feature extraction, underperformed expectations. This highlights a critical insight into Hete-CD tasks: relying solely on feature extraction proves inadequate for accurately identifying changing targets. Such intricate models often overfit to the features of a single modality, introducing bias when processing others, especially subtle changes. Similarly, complex Homo-CD methods, characterized by extensive parameters and deep architectures, are susceptible to overfitting on limited datasets, leading to reduced test set performance and compromised generalization. Furthermore, the excessive incorporation of low-level and high-level features can result in redundancy, hindering the efficient extraction of meaningful information and ultimately diminishing detection accuracy.

In contrast, high-frequency guidance-based networks offer distinct advantages in processing heterogeneous images. Rather than depending on deep feature extraction, these methods emphasize effective fusion of features across modalities. This strategy mitigates the limitations of CD approaches in cross-modal detection, delivering enhanced adaptability and accuracy for Hete-CD tasks.

3) Performance Analysis: In this section, we present a comparative analysis of distinct deep learning models across three datasets—U.K.-HCD, France-HCD, and GF-HCD—assessing both their CD performance and computational complexity. Key metrics include model parameters (Params, M), floating-point operations (FLOPs, G), $F1$-score, and IoU, with comprehensive results detailed in Table V. Parameter counts and FLOPs are further visualized in Fig. 10.

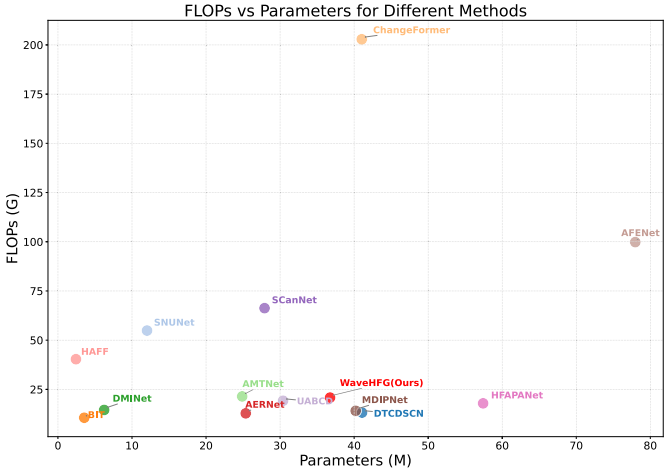


Fig. 10. Comparison of the complexity of our method with the SOTA methods.

Among models with substantial computational demands, HFA-PANet (77.97 M) and AFENet (57.42 M) stand out, alongside DTCDSCN (41.07 M), UAB-CD (30.40 M) and MDIPNet (40.21 M). Moderate-sized options like AMTNet (24.88 M) balance performance and complexity. Conversely, BIT (3.55 M, 10.59 G) and HAFF (2.43 M, 40.33 G), despite minimal parameters, exhibit a clear accuracy tradeoff.

Focusing on Hete-CD specific methods, our WaveHFG, with 36.75 M parameters, maintains a moderate complexity, particularly when compared to the parameter-heavy Hete-CD models like HFA-PANet and AFENet. Beyond its superior accuracy, WaveHFG demonstrates notable computational efficiency with FLOPs of 20.83 G. This is relatively low among high-performing models and significantly more efficient

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TABLE V
COMPARISON OF SOTA METHODS ON THREE DATASETS (U.K.-HCD, FRANCE-HCD, GF-HCD)

Model	Params	FLOPs	UK-HCD		France-HCD		GF-HCD	
	(M)	(G)	F1	IoU	F1	IoU	F1	IoU
DTCDSCN	41.07	13.21	69.60	53.42	42.17	26.72	52.55	35.64
SNUNet	12.03	54.88	72.71	57.12	53.05	36.10	62.08	45.01
BIT	3.55	10.59	43.04	27.42	43.03	27.42	44.27	28.43
ChangeFormer	41.03	202.87	87.37	77.57	37.83	23.33	64.96	48.11
DMINet	6.24	14.55	64.19	47.26	37.07	22.76	58.32	41.17
AMTNet	24.88	21.42	80.04	66.72	38.32	23.70	62.49	45.44
AERNet	25.36	12.82	68.09	51.62	47.03	30.75	59.16	42.01
SCanNet	27.90	66.27	85.37	74.47	50.01	33.35	53.09	36.14
MDIPNet	40.21	14.10	83.84	72.17	46.86	30.60	52.03	35.16
UAB-CD	30.40	19.22	77.58	63.37	46.53	30.32	65.77	49.00
HAFF	2.43	40.33	75.81	61.05	41.43	26.13	49.74	33.10
AFENet	77.97	99.82	89.80	81.48	46.28	30.11	64.86	48.00
HFA-PANet	57.42	17.90	87.07	77.11	47.61	31.24	66.05	49.43
WaveHFG	36.75	20.83	90.06	81.92	59.08	41.95	66.75	50.10

than AFENet, while being comparable to HFA-PANet in FLOPs despite HFA-PANet's larger parameter count. This highlights WaveHFG's effective optimization of computational resources. Overall, WaveHFG strikes an optimal balance between exceptional detection performance across diverse datasets and commendable computational efficiency, outperforming other specialized Hete-CD methods like HAFF, AFENet, and HFA-PANet in key accuracy metrics while maintaining a competitive or superior efficiency profile.

V. DISCUSSION

Our parameter analysis in Section IV-C reveals that the optimal weighting α for the high-frequency loss L_{HF} varies with dataset characteristics. This suggests that larger and more diverse datasets may benefit from stronger structural guidance to learn modality-invariant features. While manual tuning is standard practice, developing a robust, adaptive parameter selection method, possibly through meta-learning or a dynamic weighting scheme, presents a promising avenue for future research.

Regarding deployment, WaveHFG's CNN-based architecture and robustness make it a strong candidate for near-real-time applications. However, for strictly real-time scenarios, the modest computational overhead from the wavelet transforms would necessitate further optimization, such as model lightweighting.

Despite its strong performance, our proposed method has limitations that also guide future work. First, the model's high sensitivity to structural details, while a key strength, can also make it susceptible to high-frequency noise like SAR speckle, potentially increasing false positives. Future research could explore noise-robust guidance mechanisms. Second, as a supervised method, WaveHFG's reliance on large labeled datasets is a practical challenge. Extending the principle of high-frequency guidance to unsupervised or semi-supervised paradigms would significantly broaden its applicability.

VI. CONCLUSION

To address the challenges of CD in HRSI, we developed a comprehensive dataset with diverse target types, varied scenes, and a larger data volume. Additionally, We also propose WaveHFG, a novel network leveraging high-frequency guidance to enhance heterogeneous image analysis. This network integrates the complementary strengths of the frequency and spatial domains feature extraction, significantly improving CD across heterogeneous images. Specifically, it employs two high-frequency strategies: Diff-G, which unifies difference features in wavelet-transformed high-frequency components using Euclidean distance for matching, and Dir-G, which enhances edge precision by exploiting directional attributes of high-frequency subbands. To account for the multidimensional properties observed in both frequency and spatial domains, we further propose FSD^{2}F module. This module employs 3-D convolution to fuse features from different domains, replacing conventional concatenation operations, hence enabling the model to extract more comprehensive and detailed difference information across these domains. Extensive experiments conducted across two publicly available datasets and our newly constructed dataset affirm the efficacy and substantial scalability of WaveHFG for Hete-CD.

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